

样本协方差: $\text{Cov}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{y}_i - \bar{\mathbf{y}}) / (n - 1)$

大样本定理.: $\text{Cov}(\mathbf{x}, \mathbf{y}) = \text{E}((\mathbf{X} - \text{E}(\mathbf{X}))(\mathbf{Y} - \text{E}(\mathbf{Y}))) / (n \rightarrow \infty)$

相关系数: $r = \text{Cov}(\mathbf{x}, \mathbf{y}) / \sqrt{(\text{Var}(\mathbf{x})\text{Var}(\mathbf{y}))} = \hat{\rho}_{\text{Sxy}} / \sqrt{(\text{S}_{\text{xx}}\text{S}_{\text{yy}})}$

In regression, one variable is considered independent variable(**predictor**); other dependent variable(**outcome**).
线性回归假设: **I**. The relationship between $\mathbf{x}$ and $\mathbf{y}$ is **linear**;
II. $\mathbf{y}$ is **distributed normally** at each value $\mathbf{x}$; **III**. The variance of $\mathbf{y}$ at every value of $\mathbf{x}$ is the same (**homogeneity**); **IV**. The observation is **independent**.

简单线性回归模型 假设: $\text{E}(\epsilon_i) = 0$; $\text{Var}(\epsilon_i) = \sigma^2$; $\text{Cov}(\epsilon_i, \epsilon_j) = 0$

最小二乘法: $\min \sum_{i=1}^n (\mathbf{y}_i - \mathbf{y}_i^{\text{hat}})^2$

$\beta_1^{\text{hat}} = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{y}_i - \bar{\mathbf{y}}) / \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^2 = \Delta_{\text{Sxy}} / \text{S}_{\text{xx}}$

$\beta_0^{\text{hat}} = \bar{\mathbf{y}} - \beta_1^{\text{hat}}\bar{\mathbf{x}}$

$S^2 = \text{SSE} / (n - 2) = \sum_{i=1}^n (\mathbf{y}_i - \mathbf{y}_i^{\text{hat}})^2 / (n - 2)$ Unbiased for σ^2

极大似然法: $S_{\text{MLE}}^2 = \text{SSE} / n = (n - 2) S^2 / n$

$\beta_1^{\text{hat}} \sim \text{N}(\beta_1, \sigma^2 / \text{S}_{\text{xx}})$, $\beta_0^{\text{hat}} \sim \text{N}(\beta_0, \sigma^2 (1/n + \bar{\mathbf{x}}^2 / \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^2))$

$(n - 2) S_{\text{SE}}^2 / \sigma^2 \sim \chi^2(n - 2)$. 考虑到 β_0^{hat} 与 S^2 独立，所以

$(\beta_0^{\text{hat}} - \beta_0) / (\text{Sv}(1/n + \bar{\mathbf{x}}^2 / \text{S}_{\text{xx}})) \sim \text{t}(n - 2)$

β_0 的 (1- α) CI: $[\beta_0^{\text{hat}} \pm S \text{t}(\alpha/2, n - 2)] / \sqrt{(1/n + \bar{\mathbf{x}}^2 / \text{S}_{\text{xx}})}$

β_1 双边 $\mathbf{t}$ 检验: $\text{H}_0: \beta_1 = 0$ vs. $\text{H}_1: \beta_1 \neq 0$

$\mathbf{t} = (\beta_1^{\text{hat}} - \beta_1) \sqrt{(\text{S}_{\text{xx}}) / S} \sim \text{t}(n - 2)$, reject H_0 if $|\mathbf{t}| > \text{t}(\alpha/2, n - 2)$

(1- α) CI: $[\beta_1^{\text{hat}} - S \text{t}(\alpha/2, n - 2) / \sqrt{\text{S}_{\text{xx}}}, \beta_1^{\text{hat}} + S \text{t}(\alpha/2, n - 2) / \sqrt{\text{S}_{\text{xx}}}]$

Est. of Mean of the Response Variable for a Given Level of $\mathbf{x}$:

$\mathbf{y}_h^{\text{hat}} \sim \text{N}(\beta_0 + \beta_1 \mathbf{x}_h, \sigma^2 [1/n + (\mathbf{x}_h - \bar{\mathbf{x}})^2 / \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^2])$ Est. of $\text{E}(\mathbf{y}_h)$

CI of $\text{E}(\mathbf{y}_h)$: $[\mathbf{y}_h^{\text{hat}} \pm \mathbf{t}(\alpha/2, n - 2) S \sqrt{(1/n + (\mathbf{x}_h - \bar{\mathbf{x}})^2 / \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^2)}]$

Prediction of a new observation

$\mathbf{y}_h - \mathbf{y}_h^{\text{hat}} \sim \text{N}(0, \sigma^2 [1 + 1/n + (\mathbf{x}_h - \bar{\mathbf{x}})^2 / \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^2])$

Predictive $\mathbf{ltv}$: $[\mathbf{y}_h \pm \mathbf{t}_{\alpha/2, n - 2} S \sqrt{(1 + 1/n + (\mathbf{x}_h - \bar{\mathbf{x}})^2 / \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})^2)}]$

Analysis of Variance Approach to Regression Analysis

SST = $\sum_i (\mathbf{y}_i - \bar{\mathbf{y}})^2 = \sum_i (\mathbf{y}_i - \mathbf{y}_i^{\text{hat}})^2 + \sum_i (\mathbf{y}_i^{\text{hat}} - \bar{\mathbf{y}})^2 = \text{SSE} + \text{SSR}$

SSE Sum of squares of Residual (Unexplained Variation)

SSR Sum of squares due to Regression (Explained Var.)

Analysis of Variation (ANOVA) Table

	SS	Df	MS	F
Regression	SSR	1	MSR=SSR	MSR/MSE
Error	SSE	n-2	MSE=SSE/(n-2)	
Total	SST	n-1		

F 检验: $\text{H}_0: \beta_1 = 0$ (No linear relationship) vs. $\text{H}_1: \beta_1 \neq 0$

Reject H_0 if $\text{F} > \text{F}(\alpha, 1, n - 2)$

Coefficient of Determination (R-squared)

$\text{R}^2 = \text{SSR} / \text{SST} = 1 - (\text{SSE} / \text{SST})$

$\mathbf{r} = \pm \sqrt{(\text{R}^2)}$ True for Simple Linear Regression only

满秩分解 **Factorization:** $\mathbf{A}_{p \times q} = \mathbf{K}_{p \times r} \mathbf{L}_{r \times q}$, $\mathbf{r} = \text{r}(\mathbf{A})$, $\mathbf{K}$, $\mathbf{L}$ full rank

Let $\mathbf{B} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$; $\mathbf{D} = \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$

Remark 3.1 A'=

$\mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{B}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}$	$-\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{B}^{-1}$
$-\mathbf{B}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}$	$\mathbf{B}^{-1}$
=	
$\mathbf{D}^{-1}$	$-\mathbf{D}^{-1} \mathbf{A}_{12} \mathbf{A}_{22}^{-1}$
$-\mathbf{A}_{22}^{-1} \mathbf{A}_{21} \mathbf{D}^{-1}$	$\mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1} \mathbf{A}_{21} \mathbf{D}^{-1} \mathbf{A}_{12} \mathbf{A}_{22}^{-1}$

(A+cc')⁻¹ = $\mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{c} \mathbf{c}' \mathbf{A}^{-1} / (1 + \mathbf{c}' \mathbf{A}^{-1} \mathbf{c})$

幂等矩阵 **Idempotent matrix** $\mathbf{A}^2 = \mathbf{A}$

对于幂等矩阵 $\mathbf{A}$ ，有 **$\mathbf{r}(\mathbf{A}) = \text{tr}(\mathbf{A})$**

证明：考虑 $\mathbf{A}$ 的满秩分解 $\mathbf{A} = \mathbf{BC}$, $\mathbf{A}^2 = \mathbf{BCBC} = \mathbf{BC}$, $\mathbf{CB} = \mathbf{I}_r$

所以 $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{BC}) = \text{tr}(\mathbf{CB}) = \mathbf{r} = \text{rank}(\mathbf{A})$.

幂等矩阵的特征值均为 0 或 1.

Theorem 秩 $\mathbf{r}$ 矩阵 $\mathbf{A}_{n \times p}$ 假如可划分为 $[[\mathbf{A}_{11} \ \mathbf{A}_{12}][\mathbf{A}_{21} \ \mathbf{A}_{22}]]$, 且 $\mathbf{A}_{11}$ 是 $\mathbf{r}$ 阶满秩矩阵, 则 $\mathbf{A}' = [[\mathbf{A}_{11}^{-1} \ 0][0 \ 0]]$
类似地, 若 $\mathbf{A}_{12}/\mathbf{A}_{21}/\mathbf{A}_{22}$ 分别是 $\mathbf{r}$ 阶满秩矩阵, 则 $\mathbf{A}'$ 分别可以是 $[[0 \ 0][\mathbf{A}_{12}^{-1} \ 0]]/[[0 \ \mathbf{A}_{21}^{-1}][0 \ 0]]/[[0 \ 0][0 \ \mathbf{A}_{22}^{-1}]]$

广义逆 **Generalized Inverse $\mathbf{A}^{\circ}$:** $\mathbf{AA}^{\circ} \mathbf{A} = \mathbf{A}$; $\text{rank}(\mathbf{A}) = k$, $\mathbf{A}_{m \times n}$
 $\text{rank}(\mathbf{A}) \geq \text{rank}(\mathbf{A}^{\circ} \mathbf{A}) = \text{rank}(\mathbf{AA}^{\circ}) = k$; $\mathbf{A}^{\circ}$ 是 $\mathbf{A}$ 的广义逆
 $\mathbf{A}^{\circ} \mathbf{A}$ 和 $\mathbf{AA}^{\circ}$ 是幂等的; $\mathbf{A}^{\circ} \mathbf{A} = \mathbf{I} \leftrightarrow \text{rank}(\mathbf{A}) = n$
 $\mathbf{AA}^{\circ} = \mathbf{I} \leftrightarrow \text{rank}(\mathbf{A}) = m$; $\text{tr}(\mathbf{A}^{\circ} \mathbf{A}) = \text{tr}(\mathbf{AA}^{\circ}) = k$
 $\mathbf{K} = \mathbf{A}(\mathbf{AA}^{\circ})^{\circ}$ $\mathbf{A}'$ 对任意 $(\mathbf{AA}^{\circ})$ 均不变
 $\mathbf{K} = \mathbf{K}^{\circ}$; $\mathbf{K} = \mathbf{K}^2$ (Symmetric, Idempotent)
 $\text{rank}(\mathbf{K}) = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{K})$; $\mathbf{KA} = \mathbf{A}$; $\mathbf{A}' \mathbf{K} = \mathbf{A}'$
 $(\mathbf{X}' \mathbf{X})^{\circ} \mathbf{X}'$ is a g-inverse of $\mathbf{X}$ for any g-inverse of $\mathbf{X}' \mathbf{X}$
 $\mathbf{X}(\mathbf{X}' \mathbf{X})^{\circ}$ is a g-inverse of $\mathbf{X}'$ for any g-inverse of $\mathbf{X}' \mathbf{X}$

Vector and Matrix Calculus:

$\mathbf{x}$ 是向量: $\mathbf{u} = \mathbf{f}(\mathbf{x})$

$\mathbf{f}(\mathbf{x}) = \mathbf{a}' \mathbf{x} = \mathbf{x}' \mathbf{a} \rightarrow \partial \mathbf{u} / \partial \mathbf{x} = \mathbf{a}$

$\mathbf{f}(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x} \rightarrow \partial \mathbf{u} / \partial \mathbf{x} = 2 \mathbf{A} \mathbf{x}$

$\mathbf{X}$ 是矩阵: $\mathbf{u} = \mathbf{f}(\mathbf{X}_{p \times p})$, $\partial \mathbf{u} / \partial \mathbf{X} = [\partial \mathbf{u} / \partial \mathbf{X}_{ij}]$

$\mathbf{f}(\mathbf{X}) = \text{tr}(\mathbf{XA}) \rightarrow \partial \mathbf{u} / \partial \mathbf{X} = \mathbf{A} + \mathbf{A}' - \text{diag}(\mathbf{A})$

$\mathbf{f}(\mathbf{X}) = \ln |\mathbf{A}| \rightarrow \partial \mathbf{u} / \partial \mathbf{X} = 2 \mathbf{X}^{-1} - \text{diag}(\mathbf{X}^{-1})$

$\mathbf{x}$ 是常数: $\mathbf{A}_{n \times n} = [\mathbf{a}_{ij}]$, $\mathbf{a}_{ij}$ 是 $\mathbf{x}$ 的函数,

$\mathbf{f}(\mathbf{x}) = \mathbf{A}^{-1} \rightarrow \partial \mathbf{u} / \partial \mathbf{x} = -\mathbf{A}^{-1} \partial \mathbf{A} / \partial \mathbf{x} \mathbf{A}^{-1}$

$\mathbf{f}(\mathbf{x}) = \ln |\mathbf{A}| \rightarrow \partial \mathbf{u} / \partial \mathbf{x} = \text{tr}(\mathbf{A}^{-1} \partial \mathbf{A} / \partial \mathbf{x})$

Random Vector and Multivariate Normal Distribution

$\text{Cov}(\mathbf{Y}) = \Sigma = \text{E}((\mathbf{Y} - \text{E}(\mathbf{Y}))(\mathbf{Y} - \text{E}(\mathbf{Y}))') = [\sigma_{ij}] = [\text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)] \geq \mathbf{0}$

$\text{Cov}(\mathbf{AY}) = \mathbf{ACov}(\mathbf{Y}) \mathbf{A}'$ $\mathbf{A}$ 是常数矩阵

$\text{Cov}(\mathbf{AX}, \mathbf{BY}) = \mathbf{ACov}(\mathbf{X}, \mathbf{Y}) \mathbf{B}'$

Mahalabalis Distance: $(\mathbf{Y} - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{Y} - \boldsymbol{\mu})$ (Standardized Dist.)
Generalized Variance = $|\Sigma|$

Remark 4.1 一系列观察 $\mathbf{y}_i = [\mathbf{y}_{i1} \ \dots \ \mathbf{y}_{ip}] \sim (\text{i.i.d.}) (\boldsymbol{\mu}, \Sigma)$

样本均值向量 $\boldsymbol{\mu}^{\text{hat}} = \bar{\mathbf{y}} = [\bar{\mathbf{y}}_1 \ \dots \ \bar{\mathbf{y}}_p]$, $\mathbf{y}_j = \sum_{i=1}^n \mathbf{y}_{ji} / n$

样本协方差矩阵 $\Sigma^{\text{hat}}_{p \times p} = \mathbf{S} = \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})'$

相关系数矩阵 $\Omega = [\rho_{ij}] = [\sigma_{ij} / \sqrt{(\sigma_{ii} \sigma_{jj})}]$

Remark 4.2 令 $\mathbf{A}$ 为对称矩阵, $\text{E}(\mathbf{YAY}) = \text{tr}(\mathbf{A} \Sigma) + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}$

证明: $\text{E}(\mathbf{YAY}) = \text{E}((\mathbf{Y} - \boldsymbol{\mu})' \mathbf{A} (\mathbf{Y} - \boldsymbol{\mu})) + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}$

$\text{E}((\mathbf{Y} - \boldsymbol{\mu})' \mathbf{A} (\mathbf{Y} - \boldsymbol{\mu})) = \text{E} \text{tr}((\mathbf{Y} - \boldsymbol{\mu})' \mathbf{A} (\mathbf{Y} - \boldsymbol{\mu})) = \text{E} \text{tr}(\mathbf{A} (\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})')$
 $= \text{tr}(\mathbf{A} \text{E}((\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})')) = \text{tr}(\mathbf{A} \Sigma)$

Theorem: $\mathbf{Y}_1, \dots, \mathbf{Y}_m$ 独立 $\rightarrow \mathbf{g}_1(\mathbf{Y}_1), \dots, \mathbf{g}_m(\mathbf{Y}_m)$ 独立.

MGF: $\mathbf{M}_{\mathbf{Y}}(\mathbf{t}) = \text{E}(\mathbf{e}^{\mathbf{t}' \mathbf{Y}})$

Let $\mathbf{Y}_{p \times 1} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, $\mathbf{M}_{\mathbf{Y}}(\mathbf{t}) = \mathbf{e}^{\mathbf{t}' \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}' \Sigma \mathbf{t}}$

$\mathbf{f}_{\mathbf{Y}}(\mathbf{y}) = |\Sigma|^{-1/2} (2\pi)^{-p/2} \exp(-(\mathbf{y} - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{y} - \boldsymbol{\mu}) / 2)$

Remark 4.4 令 $\mathbf{B}$ 为常数矩阵, $\mathbf{C}$ 为常数向量, 则

$\mathbf{X} = \mathbf{BY} + \mathbf{C} \sim \text{N}(\mathbf{B} \boldsymbol{\mu} + \mathbf{C}, \mathbf{B} \Sigma \mathbf{B}')$

证明: $\mathbf{M}_{\mathbf{X}}(\mathbf{t}) = \text{E}(\mathbf{e}^{\mathbf{t}' (\mathbf{BY} + \mathbf{C})}) = \text{E}(\mathbf{e}^{\mathbf{t}' \mathbf{B} \mathbf{Y}}) \mathbf{e}^{\mathbf{t}' \mathbf{C}}$ 记 $\mathbf{t}^* = \mathbf{B}' \mathbf{t}$, 则

$\mathbf{M}_{\mathbf{X}}(\mathbf{t}) = \text{E}(\mathbf{e}^{\mathbf{t}^{*'} \mathbf{Y}}) \mathbf{e}^{\mathbf{t}' \mathbf{C}} = \mathbf{M}_{\mathbf{Y}}(\mathbf{t}^*) \mathbf{e}^{\mathbf{t}' \mathbf{C}} = \exp(\mathbf{t}^{*'} \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^{*'} \Sigma \mathbf{t}^* + \mathbf{t}' \mathbf{C})$
 $= \exp(\mathbf{t}' (\mathbf{B} \boldsymbol{\mu} + \mathbf{C}) + \frac{1}{2} \mathbf{t}' \mathbf{B} \Sigma \mathbf{B}' \mathbf{t})$. 即证.

令 $\mathbf{Y} = [[\mathbf{Y}_1] \ [\mathbf{Y}_2]] \sim \text{N}([[\boldsymbol{\mu}_1] [\boldsymbol{\mu}_2]], [[\Sigma_{11} \ \Sigma_{12}][\Sigma_{21} \ \Sigma_{22}]]$,
 $\mathbf{Y}_1 \sim \text{N}(\boldsymbol{\mu}_1, \Sigma_{11})$
 $\mathbf{Y}_1 | (\mathbf{Y}_2 = \mathbf{y}_2) \sim \text{N}(\boldsymbol{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})$
 $\mathbf{Y}_1$ 和 $\mathbf{Y}_2$ 独立当且仅当 $\Sigma_{12} = 0$

偏相关 **Partial Correlation:**

$\mathbf{v} = [|\mathbf{y}| |\mathbf{x}|] \sim \text{N}_q([[\boldsymbol{\mu}_1] [\boldsymbol{\mu}_2]], [[\Sigma_{\mathbf{Y}\mathbf{Y}} \ \Sigma_{\mathbf{Y}\mathbf{X}}][\Sigma_{\mathbf{X}\mathbf{Y}} \ \Sigma_{\mathbf{X}\mathbf{X}}]])$

$\Sigma_{\mathbf{Y}, \mathbf{X}} = \Sigma_{\mathbf{Y}\mathbf{Y}} \Sigma_{\mathbf{Y}\mathbf{X}} \Sigma_{\mathbf{X}\mathbf{X}}^{-1} \Sigma_{\mathbf{X}\mathbf{Y}} = [\sigma_{ij}]_{i,j,q}$, $\mathbf{D}_{\mathbf{Y}, \mathbf{X}} = [\text{diag}(\Sigma_{\mathbf{Y}, \mathbf{X}})]^{1/2}$

$\rho_{ij, r-q} = \sigma_{ij, r-q} / \sqrt{(\sigma_{ii, r-q} \sigma_{jj, r-q})}$, $\Omega_{\mathbf{Y}, \mathbf{X}} = \mathbf{D}_{\mathbf{Y}, \mathbf{X}}^{-1} \Sigma_{\mathbf{Y}, \mathbf{X}} \mathbf{D}_{\mathbf{Y}, \mathbf{X}}^{-1}$

二次型 Quadratic Forms: 令 $\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, $\mathbf{A}$ 是对称矩阵, 则

$\mathbf{M}_{\mathbf{x}' \mathbf{A} \mathbf{x}}(\mathbf{t}) = | \mathbf{I} - 2 \mathbf{tA} \Sigma |^{-1/2} \exp(-\boldsymbol{\mu}' (\mathbf{I} - (\mathbf{I} - 2 \mathbf{tA} \Sigma)^{-1})^{-1} \Sigma^{-1} \boldsymbol{\mu} / 2)$

$\text{E}(\mathbf{x}' \mathbf{A} \mathbf{x}) = \text{tr}(\mathbf{A} \Sigma) + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}$, $\text{Var}(\mathbf{x}' \mathbf{A} \mathbf{x}) = 2 \text{tr}[(\mathbf{A} \Sigma)^2] + 4 \boldsymbol{\mu}' \mathbf{A} \Sigma \mathbf{A} \boldsymbol{\mu}$

Non-central χ^2 distribution

$\mathbf{x} \sim \text{N}(\mathbf{0}, \mathbf{I}_n)$, $\mathbf{x}' \mathbf{x} \sim \chi^2(n)$; $\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \mathbf{I}_n)$, $\mathbf{u} = \mathbf{x}' \mathbf{x} \sim \chi^2(n, \lambda)$, $\lambda = \boldsymbol{\mu}' \boldsymbol{\mu} / 2$

$\mathbf{f}(\mathbf{u}) = \mathbf{e}^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbf{u}^{n/2+k-1} \mathbf{e}^{-\mathbf{u}/2} / (k! 2^{n/2+k} \Gamma(n/2+k))$, $\mu > 0$, $\lambda \geq 0$

$\mathbf{M}_{\mathbf{u}}(\mathbf{t}) = (1 - 2\mathbf{t})^{-n/2} \exp(-\lambda [1 - (1 - 2\mathbf{t})^{-1}])$; $\text{E}(\mathbf{u}) = n + 2\lambda$,

$\text{Var}(\mathbf{u}) = 2n + 8\lambda$; $\mathbf{u} \sim \chi^2(n, \lambda)$ 独立, $\sum \mathbf{u}_i \sim \chi^2(\sum n_i, \sum \lambda_i)$

Non-central F distribution

$\mathbf{u}_1 \sim \chi^2(p_1, \lambda)$, $\mathbf{u}_2 \sim \chi^2(p_2, 0)$, $\mathbf{w} = \mathbf{u}_1 p_2 / \mathbf{u}_2 p_1 \sim \text{F}(p_1, p_2, \lambda)$

$\text{E}(\mathbf{w}) = (1 + 2\lambda / p_1) p_2 / (p_2 - 2)$

Non-central t distribution

$\mathbf{z} \sim \text{N}(\boldsymbol{\mu}, 1)$, $\mathbf{u} \sim \chi^2(n)$, $\mathbf{z}$ is independent of $\mathbf{u}$, then

$\mathbf{t} = \mathbf{z} / \sqrt{(\mathbf{u} / n)} \sim \text{non-centered t distribution}$

Theorem 5.1 $\mathbf{x}_{p \times 1} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, $\mathbf{A}$ 对称, $\mathbf{r} = \text{rank}(\mathbf{A})$, $\lambda = \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu} / 2$ 则

$\mathbf{q} = \mathbf{x}' \mathbf{A} \mathbf{x} \sim \chi^2(\mathbf{r}, \lambda) \leftrightarrow \mathbf{A} \Sigma$ 幂等.

推论

$\mathbf{x} \sim \text{N}(\mathbf{0}, \mathbf{I})$, 则 $\mathbf{x}' \mathbf{A} \mathbf{x} \sim \chi^2(\mathbf{r}) \leftrightarrow \mathbf{A}$ 幂等

$\mathbf{x} \sim \text{N}(\mathbf{0}, \Sigma)$, 则 $\mathbf{x}' \mathbf{A} \mathbf{x} \sim \chi^2(\mathbf{r}) \leftrightarrow \mathbf{A} \Sigma$ 幂等

$\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, then $\mathbf{x}' \mathbf{x} / \sigma^2 \sim \chi^2(n, \boldsymbol{\mu}' \boldsymbol{\mu} / (2\sigma^2))$

$\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \mathbf{I})$, 则 $\mathbf{x}' \mathbf{A} \mathbf{x} \sim \chi^2(\mathbf{r}, \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu} / 2) \leftrightarrow \mathbf{A}$ 幂等

Theorem 5.2

$\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, $\mathbf{A}$ 对称, 则 $\mathbf{x}' \mathbf{A} \mathbf{x}$ 与 $\mathbf{B} \mathbf{x}$ 独立 $\leftrightarrow \mathbf{B} \Sigma \mathbf{A} = \mathbf{0}$.

Theorem 5.3

$\mathbf{x} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, $\mathbf{A}$ 和 $\mathbf{B}$ 对称, $\mathbf{r} = \text{rank}(\mathbf{A})$, 则

$\mathbf{x}' \mathbf{A} \mathbf{x}$ 和 $\mathbf{x}' \mathbf{B} \mathbf{x}$ 均幂等 $\leftrightarrow \mathbf{A} \Sigma \mathbf{B} = \mathbf{B} \Sigma \mathbf{A} = \mathbf{0}$.

Remark 4.3 $\mathbf{Y}_{p \times 1} \sim \text{N}(\boldsymbol{\mu}, \Sigma)$, 则 $\mathbf{M}_{\mathbf{Y}}(\mathbf{t}) = \mathbf{e}^{\mathbf{t}' \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}' \Sigma \mathbf{t}}$.

模型误用 Model Misspecification

误差结构: $\mathbf{y}=\mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon}$ T: $\text{Cov}(\mathbf{y})=\sigma^2\mathbf{I}$, F: $\text{Cov}(\mathbf{y})=\sigma^2\mathbf{I}$
E($\boldsymbol{\beta}^{**}$)= $\boldsymbol{\beta}$ Unbiased. $\text{Var}(\boldsymbol{\beta}^{**})=(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\geq\sigma^2\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}$
均值:
欠拟合 Under-fit: T: $\mathbf{y}=[\mathbf{X}_1\ \mathbf{X}_2][\boldsymbol{\beta}_1'\ \boldsymbol{\beta}_2']'+\boldsymbol{\varepsilon}$, F: $\mathbf{y}=\mathbf{X}_1\boldsymbol{\beta}_1+\boldsymbol{\varepsilon}$
 $\boldsymbol{\beta}_1^{**}=(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$. ($\boldsymbol{\beta}_1\in\mathbb{R}^{(p+1)}$, $\boldsymbol{\beta}_2\in\mathbb{R}^{(k-p)}$)
E($\boldsymbol{\beta}_1^{**}$)= $\boldsymbol{\beta}_1+(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2$. $\text{Cov}(\boldsymbol{\beta}_1^{**})=\sigma^2(\mathbf{X}_1'\mathbf{X}_1)^{-1}$.
 $\text{S}^2=(\mathbf{y}-\mathbf{X}_1\boldsymbol{\beta}_1^{**})'(\mathbf{y}-\mathbf{X}_1\boldsymbol{\beta}_1^{**})/(n-p-1)$
E(S_1^2)= $\sigma^2+\boldsymbol{\beta}_2'\mathbf{X}_2'(1-\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1')\mathbf{X}_2\boldsymbol{\beta}_2$.
E($\mathbf{y}^{**}$)=E($\mathbf{y}^{*}$)+(x2*-x1*(X1'X1)⁻¹X1'X2)β2. All biased.
过拟合 Over-fit: T: $\mathbf{y}=\mathbf{X}_1\boldsymbol{\beta}_1+\boldsymbol{\varepsilon}$, F: $\mathbf{y}=[\mathbf{X}_1\ \mathbf{X}_2][\boldsymbol{\beta}_1'\ \boldsymbol{\beta}_2']'+\boldsymbol{\varepsilon}$
E($\boldsymbol{\beta}_1^{**}$)= $\boldsymbol{\beta}_1$, E($\boldsymbol{\beta}_2^{**}$)= $\mathbf{0}$, Unbiased. $\text{Var}(\boldsymbol{\beta}^{**})=\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$
 $\text{Var}(\boldsymbol{\beta}_1^{**})=\sigma^2(\mathbf{X}_1'\mathbf{X}_1)^{-1}(1+\text{CBC}'(\mathbf{X}_1'\mathbf{X}_1)^{-1})\geq\text{Var}(\boldsymbol{\beta}_1^{*})$.

假设检验 Hypothesis Testing

H0: $\mathbf{K}'\boldsymbol{\beta}=\boldsymbol{\mu}$. $\mathbf{K}'\in\mathbb{R}^{q\times(p+1)}$, 行满秩
Q=($\mathbf{K}'\boldsymbol{\beta}^{*}-\boldsymbol{\mu}$)'($\mathbf{K}'(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{K}$)^{-1}($\mathbf{K}'\boldsymbol{\beta}^{*}-\boldsymbol{\mu}$)=SSH
F(H)=(Q/q)/(SSE/(n-r(X)))=Q/(qσ²~)-F(q, n-r(X),
($\mathbf{K}\boldsymbol{\beta}-\boldsymbol{\mu}$)'(K'(X'X)⁻¹K)⁻¹(K'β-μ)/(2σ²))~|h0 F(q,n-r(X))
p-value=P(F(q,n-r(X))>F(H))
K'β*-μ~N(K'β-μ, K'(X'X)⁻¹Kσ²).
Q/σ²~χ²(q, (K'β-μ)'(K'(X'V⁻¹X)⁻¹K)⁻¹(K'β-μ)/(2σ²))

考虑在 K'a=μ限制下的模型 y~N(Xa, σ²I)
min (y-Xa)'(y-Xa) + 2λ(K'a-μ)
a*=β^-(X'X)⁻¹K(K'(X'X)⁻¹K)⁻¹(K'β*-μ).是模型的 b.l.u.e.

SSE=(y-Xβ^{*})'V⁻¹(y-Xβ^{*})~N(0,
SSEH0=(y-Xa^{*})'(y-Xa^{*})=SSE+(β^{*}-a^{*})'X'X(β^{*}-a^{*})
=SSE+Q≥SSE.

特殊情况 Special Cases

(1) H0: β=β0.
Q=(β^{*}-β0)'X'X(β^{*}-β0), F(H)=Q/((p+1)σ²~)
F(H)~F(p+1, n-p-1), a*=β0.
(2) H0: λ'β=m.
Q=(λ'β^{*}-m)²/(λ'(X'X)⁻¹λ), F(H)=Q/(σ²~)~h0 F(1, n-r(X))
√F(H)~t(n-r(X))
a*=β^{*}-(λ'β^{*}-μ)(X'X)⁻¹λ/(λ'(X'X)⁻¹λ)
(3) H0: β=[|β1|_{(p+1-h)*1} | β2|_{h*1}=0]]
K'h^{*(p+1)}=[0|h^{*(p+1-h)} | Ih]→K'β=0.
(K'(X'X)⁻¹K)⁻¹=X2'X2-X2'X1(X1'X1)⁻¹X1'X2=X2'(I-H1)X2=B
β2^{*}=B⁻¹X2'(I-H1)y
Q=(K'β^{*}-μ)'(K'(X'X)⁻¹K)⁻¹(K'β^{*}-μ)=β2^{*}'X2'(I-H1)X2β2^{*}
Q/σ²~χ²(h, β2'X2'(I-H1)X2β2/(2σ²))
F=Q/(hσ²~)~|h0 F(h, n-p-1)

Likelihood Ratio Test LRT 似然比检验

设 y~Nn(xa, σ²I), r(X)=p+1,则假设 H0: a=0 的似然比检验
F=a~x'y/(p+1)/((y'y-a~x'y)/(n-p-1)).
F>F(α, p+1, n-p-1)时拒绝 H0.

置信区间/区域

β: (β^{*}-β)'X'X(β^{*}-β)≤(p+1)σ² F(α, p+1, n-p-1).
β: βi'±t(α/2, n-p-1)S{βi} (S{βi}'=√(σ²^(X'X)⁻¹))
λβ: λ'β'±t(α/2, n-p-1)σ^{*}√(λ'(X'X)⁻¹λ)
E(y*)|x=x*: x'β'±t(α/2, n-p-1)σ^{*}√(x'(X'X)⁻¹x)
Var(y*)=(x'(X'X)⁻¹x)σ²^
Prediction: x'β'±t(α/2, n-p-1)σ^{*}√(1+x'(X'X)⁻¹x)
σ²: [(n-p-1)σ²~/χ²(α/2, n-p-1), (n-p-1)σ²~/χ²(1-α/2, n-p-1)]

模型诊断

残差 e^{*}=(I-H)y=(I-H)ε=(I-X(X'X)⁻¹X')ε.
记 H=X(X'X)⁻¹X'=[hij], 则εi^{*}=εi-∑j=1ⁿhijεj~N(0, σ²(1-hii))
残差的性质
E(ε^{*})=0: Cov(ε^{*})=σ²(I-H)=Cov(ε^{*}, y), Cov(ε^{*}, Y^{*})=0.
ε^{*}=ε^{*}'1/n=0; ε^{*}'y=SSE; ε^{*}'Y=0; ε^{*}'X=0'
H=X(X'X)⁻¹X'的性质
1/n≤hii≤1; -1/2≤hij≤1/2
hii=1/n+(Xi1-Xi')'(Xc'Xc)⁻¹(X1i-Xi').
tr(H)=∑hii=rank(H)=p+1
中心化模型 y=[1 Xc][[α][β1]]+ε, β1=[β1... βp], Xc=(I-1/n)X1
α^{*}=y', β1=(Xc'Xc)⁻¹Xc'y, y^{*}=(1/n+Hc)y=Hy

残差分析 (记σ^{*}=S=SSE/(n-p-1))

对于残差的方差并非定值的情况，有
Studentized Residual r1=ε1^{*}/(σ^{*}√(1-hii)) (rstandard, |r1|>2)
Deleted Residual ε0^{*}=yi-yi^{*}=εi^{*}/(1-hii)
β0^{*}=β^{*}-εi^{*}(X'X)⁻¹x1/(1-hii), SSE0=SSE-ε0^{*}
y0^{*}=x1β0^{*}(不使用第1个样本得到的β的估计)
S0²=SSE0/(n-p-2)=∑i=1ⁿ(yi-yj0^{*})²/(n-1-(p+1))
S-Deleted R- ti=εi^{*}/(S0√(1-hii)) (rstudent)
PreSS=∑i=1ⁿε0^{*}2^{*}=∑i=1ⁿ(εi^{*}/(1-hii))²
强影响点 Influential Observations
Leverage hi: hii≥2(p+1)/n (hatvalues)
Cook's Distance: Di=(β0^{*}-β^{*})'X'X(β0^{*}-β^{*})/((p+1)σ²~)
Di≥F(0.5, p+1, n-p-1)
DFFITS: (yi^{*}-y0^{*})/(σ0^{*}√hii), y0^{*}=x1β0^{*}≥2√((p+1)/n)

多重共线性 Multicollinearity

检测: 方差膨胀因子 Variance Inflation Factor VIF
(VIF)=1/(1-Rj²), j = 1, ..., p. Rj 是将第 j 个自变量当作因变量
E(Xi)=α0+α1Xi+...+αi-1Xi-1+αi+1Xi+1+...+αpXp 的模型的 COD.

(VIF)≥10,即 Rj²≥0.9 时认为此变量重要。

令 X'X 的特征值为λ1≥λ2≥...≥λr. (r=p+1)

Condition Number: κ=√(λ1/λr)
Condition Indexes: √(λ1/λj), j=2, 3, ..., r
Multicollinear Problem: Condition Number≥30.

变量选择 Variable Selection

目标: 寻找由独立变量构成的最优的集合关联的模型。
要求: 移除无关变量后更容易解释, 共线性概率降低。
方法: Stepwise Regression/Best-subset/LASSO...

Elastic Net: min(y-Xβ)'(y-Xβ)+λ1∑i=1^p|βi|+λ2∑i=1^pβi²
Qualitative Independeng Variables 离散自变量

引入 Dummy Variable
Interaction Regression Models 交互回归模型 X3=X1X2
Polynomial- 多项式回归模型 yi=β0+β1xi+β2xi²+εi,
xi=Xi-X'
Coef. of Partial Determination
R²yj(Other var)=(SSEreduced-SSEfull)/SSEreduced

不满秩的/降秩模型 (X'X 不可逆, 记 G=(X'X)⁻)
OLSE: b⁰=GX'y, 不唯一

E(b⁰)=GX'XbΔ=Ab(Ab 的无偏估计, 不是 b 的)
Var(b⁰)=GX'GX'σ²; y^{*}=XGX'y. (XGX' 不变); E(y^{*})=Xb.
SSE=y'(I-XGX')y. SST=y'(I-11'/n)y
E(SSE)=σ²(n-r(X)) (-σ²=SSE/(n-r(X)) 无偏)
SSR=SST-SSE=y'(XGX'-I/n)y.

y~N(Xb, σ²I); b⁰=GX'y~N(GX'Xb, GX'XG'σ²). b⁰ 独立于 σ²^.
SSE/σ²~χ²(n-r(X)), SSR/σ²~χ²(r(X)-1, b'X'(I-1/n)Xb/(2σ²))
SSE 与 SSR 独立
F(R)=(SSR/(r(X)-1)/(SSE/(n-r(X)))~
F(r(X)-1, n-r(X), b'X'(I-11'/n)Xb/(2σ²))

Remedy over-parametrization yij=μ+αi+εij
(1) yij=αi+εij; (2) yij=μ+αi+εij
(3) α1=0, αi=Difference between ith group and 1st group.

可估函数 Estimable Functions

如果存在 t 使得 E(ty)=q'b, ∀b, 则 q'b 被称作可估的。其
中 y=Xb+ε是降秩的。如果 q'b 可估, 则存在 t 使得 t'X=q'.

可估函数 q'b 的 b.l.u.e 是 q'b⁰.
q'b⁰=q'GX'y; q'b⁰~N(q'b, q'Gqσ²) (最小方差)

定理 9.1 当且仅当 q'A=q'时, q'b 可估, 其中 A=GX'X

可测假设 Testable Hypothesis H0: K'a=μ. K∈R^{r*(p+1)}, K'a 可估
a⁰=(X'X)⁻X'y, 则 E(K'a⁰)=K'a, Var(K'a⁰)=K'GKσ²
y~N(Xa, σ²I), a⁰~N(GX'Xa, GX'XG'σ²)
K'a⁰-μ~N(K'a⁰-μ, K'GKσ²).
Q=(K'a⁰-μ)'(K'GK)⁻¹(K'a⁰-μ),
Q/σ²~χ²(r, (K'a⁰-μ)'(K'(X'X)⁻¹K)⁻¹(K'a⁰-μ)/(2σ²))
F(H)=Q/r/(SSE/(n-r(X)))~F(r, n-r(X)),
(K'a⁰-μ)'(K'GK)⁻¹(K'a⁰-μ)/(2σ²)~|h0 F(r, n-r(X))

一因素布局 (仅考虑一个变量) One-way Layout

y=Xa+ε, ε~N(0, σ²I), (yij=μ+ai+εij).
X'y=[∑i1 yij ∑i2 yij ... ∑in yij], a 不可估
a⁰=GX'y=[0 y1' y2' ... ym']
X⁰'X'y=∑i=1^myⁱ2/ni.
SSR=∑y=1^myⁱ2/ni - ny². SST=∑i=1^m∑j=1^{nj}yij² - ny².
SSE=∑i=1^m∑j=1^{nj}yij² - ∑y=1^mnyⁱ2/ni

	SS	D.f.	MS	F
Reg.	Q	r	Q/r	Q/r/(SSE/(n-r(X)))
Res.	SSE	n-r(X)	SSE/(n-r(X))	

F~|h0 F(r, n-r(X)) (H0: a1=...=am=0)

-假设检验

H0:a1=...=am, 则 K=[0 0 1 -1 ... 0] [0 1 0 -1 ...0] ... [0 1 ... 0 -1]]
Q=(K'a⁰-μ)'(K'(X'X)⁻¹K)⁻¹(K'a⁰-μ),
F=Q/(m-1)/(SSE/(n-m)) ~ F(m-1, n-m) (r=m-1, r(X)=m)

If A, B and A+PBQ are nonsingular, then
(A+PBQ)⁻¹=A⁻¹-A⁻¹PB(B+BQA⁻¹PB)⁻¹BQA⁻¹

例 X~N3(μ, Σ), μ'=[3 -2 0], X'=[X1 X2 X3],
Σ=[[5 0 -3] [0 9 0] [-3 0 2]]
a. X2 和 2X1-X3 是否独立? b. 找到(2X1-5X3, X1+X2)的分布.
c. 给定 X1=1, X2=-2, 求 X3 的条件分布.
解: a. 令 B=[[0 1 0][2 0 -1]], 则 BX=[[X2][2X1-X3]].
BX~N(Bμ, BΣB')=N([-2 6]', [[9 0][0 34]]).
注意到 Cov(X2, 2X1-X3)=0, 所以它们独立.
b. 令 C=[[2 0 -5][1 1 0]], 则 BX=[[2X1-5X3][X1+X2]].
BX~N([6 1]', [[130 25][25 14]]).
c. X3|X1=1, X2=-2 ~N(0+[-3 -0][[5 0][0 9]]⁻¹[(1 2)-[3 -2]']
,[[2]-[-3 -0]][[5 0][0 9]]⁻¹[-3 -0]')
=N((6/5, 1/5).

例 Let y be the aggregate personal saving divided by disposable income, and x1, x2, x3, x4 be percentage of population under 15, disposable income per capita (in 100 US dollars), percentage of population over 75 and the percentage rate of change in per capita disposable income respectively. We first use a multiple regression model (errors assumed to be distributed as independent N(0, σ²)) with covariates x3 and x4. Based on a dataset collected from 50 countries, we have the following R outcome.

Call: lm(formula = y ~ x3 + x4)
Coef.: Est. Std. t Pr(>|t|)
(Intercept) 5.4695 1.4101 3.879 0.000325***
x3 1.0726 ? 2.351 0.022992*
x4 0.4636 0.2052 2.259 0.028562*

Residual standard error: 4.122 on 47 degrees of freedom
Multiple R-Squared: 0.1883, Adjusted R-squared: 0.1538
F-statistic: ? on ? and ? DF, p-value: ?

(X'X)⁻¹: (Intercept) x3 x4
(Intercept) 0.1170 -0.02758 -0.008997
x3 -0.02758 0.01226 -0.0001396
x4 -0.008997 -0.0001396 0.002479

(a) 求 x3 的系数估计, 计算方差并构建 95% CI(use α=0.05, t(0.025, 47)=2.011741, t(0.05, 47)=1.677927, t(0.025, 49)=2.00975, t(0.05, 49)=1.676551).
Answer: E(β3^{*})=1.0726; σ^{*}=4.122; Var(β3^{*})=4.122²*0.01226;
CI: [β3^{*}±t(α/2, 50 - 2 - 1)·√Var(β3^{*})]=(0.1545, 1.9907)
(b) Using the fact that SST = 983.63, SSE = 798.40, conduct a test of overall model utility (α=0.05, F(0.025, 2, 47)=3.994171, F(0.05,2,47)=3.195056, F(0.025, 2, 45)=4.008502, F(0.05, 2, 45)=3.204317).
Answer: SSR=SST-SSE=185.23,
MSR=SSR/(3-1)=92.615, MSE=SSE/(50-3)=16.987,
F=92.615/16.987, F=5.452>F(0.05, 2, 47)=3.195, Reject H0.
(c) Let y[^]* be the predicted value when x3=1.06, x4=3.08, find y[^]* and its variance.
Answer: y[^]*=8.0343, Var(y[^] *)=x^{*}*(X'X)⁻¹x^{*}σ²^=0.6711.
(d) Determine the 95% confidence interval for the mean value of y when x3 =1.06, x4=3.08.
Answer: (6.3863, 9.6823).
(e) Write down the details of the test on whether we need to include covariates x1, x2 in the model. You may use the following results.

>g1=lm(y~x3+x4+x1+x2)
>anova(g2, g1)

	Res. Df.	RSS	Df.	Sum of Sq.	F	Pr(>F)
1	47	798.40				
2	45	650.71	?(2)	?(Q)	?	0.01003 *

Answer: Q = SSEH0-SSE =RSSH0-RSS= 147.69;
F=Q/2/(SSE/(n-4-1))=5.1068; p-value = Pr(F(2, 45)≥5.1068) = 0.01003
例 Consider the model yij=μ+τi+εij, i = 1,2,3, j = 1,2.

(a) Write X, X'X, X'y, and the normal equations;
Answer:
$$X=\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, X'X=\begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix}, K=\begin{bmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}, a=\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}, \tilde{\mu}=\begin{bmatrix} 0 \end{bmatrix}$$

Normal equation: X'Xa=X'y.
(b) 适当重参数化使估计唯一，并求解
Answer: 令μ=0
(c) Show that H0: τ1=τ2=τ3 are testable.
(d) Derive the detailed test procedure to test the above H0.
例 考虑模型: Yi=β0+β1Xi+εi, i = 1, 2, ..., n.其中εi ~iid. N(0, σ²). 令 Y^{*} for Y1, ..., Yn 的均值. 设β0^{*}和β1^{*}是β0和β1 (≠0)的最小二乘估计. σ²是σ²的无偏估计.
(a) 令δ=(β0+β1X)/β1, X是 X1,...,Xn 的均值,求 Y^{*} - δβ1^{*} 的分布.
Answer: β1^{*}~N(β1, σ²/Sxx). Y^{*}~N(β0+β1X^{*}, σ²/n).
→Y-δβ1^{*}~N(0, σ²(1/n + δ²/Sxx)).
(b) 求(Y-δβ1^{*})²/σ² 的分布.
Answer: F(1, n-2).

(c) 若 H0: β1=0 被一个双边α水平的 t 检验拒绝. 写出得到-β0/β1 的 1-α置信区间过程.
Answer: From (b), we have
Pr((y^{*}-δβ1^{*})²/(σ²(1/n+δ²/Sxx)) ≤Fα,1,n-2) = 1-α
→Pr(*≤0) = 1-α. * = 0 has 2 solutions of δ: say g1 and g2.
Given that H0: β1=0 was rejected at level α→β1^{*}≥-F(α, 1, n-2)σ²/Sxx≥0
Thus, Pr(g1≤δ≤g2) = Pr(g1≤(-(β0/β1+X^{*})≤g2) = 1-α
100(1-α)% CI: (X^{*}-g2, X^{*}-g1).

Simultaneous Intervals
Bonferroni CI: m intervals with αc=αt/m
For m linear functions λ1'β, ..., λm'β, 100(1-α)% CI:
λi'β^{*} ± t(α/2, m, n-p-1)σ^{*}√(λi'(X'X)⁻¹λi)
Scheffe CI: For all possible linear functions λ'β, 100(1-α)% CI:
λ'β^{*} ± σ^{*}√((p+1)F(α, p+1, n-p-1)λ'(X'X)⁻¹λ)